

Chapter 06 Homework — Ruler-and-compass constructions

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Problem 06.2. *Suppose that the point $(\ell, 0)$ is constructible (where $\ell > 0$). Show how to construct the points $(\sqrt{\ell}, 0)$ and $(\ell^2, 0)$.*

Problem 06.5. *Show that the point $(2^{1/3}, 0)$ is not constructible (impossibility of ‘duplicating the cube’).*

Problem 06.6. *Show that it is not possible to construct (a) a regular nonagon or (b) a regular heptagon, using ruler and compasses.*

Problem 06.7. (a) *Suppose that x and y are real numbers such that $[\mathbb{Q}(x, y) : \mathbb{Q}] = 2$. Show that (x, y) is constructible.*

(b) *Suppose that x and y are real numbers, and that $\mathbb{Q} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_r = \mathbb{Q}(x, y)$ is an increasing sequence of fields such that $[F_{j+1} : F_j] = 2$ for $0 \leq j < r$. Show, by induction on r , that (x, y) is constructible.*